



Statistical Natural Language Processing Term Project Report

Image Captioning with Encoder-Decoder Architectures on MS-COCO

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<https://drive.google.com/drive/folders/1nTiCsNWdTLWZTdeN0H1VssbWVN426VD?usp=sharing>

1. Introduction

This project aims to develop an "Image Captioning" system capable of describing image contents with natural language using the MS-COCO dataset. Within the scope of the project, starting from the basic Encoder-Decoder structure, the Teacher Forcing technique, various Global Attention mechanisms, and Region Attention networks were integrated into the system step-by-step, and the performance results were analyzed comparatively.

2. Methodology and System Architecture

2.1. Dataset and Data Splitting Strategy

The models were trained and evaluated using a carefully selected subset of the MS-COCO 2014 training dataset. To ensure computational efficiency while maintaining diversity, a total of 30,000 images and their corresponding captions were utilized. This subset was split into 80% for training (24,000 images) and 20% for validation (6,000 images). For the final quantitative evaluation and comparison of all 5 models (Phase 1, 2, and 3), a completely unseen test set of 500 images was randomly sampled from the validation split.

2.2. Encoder-Decoder Foundation and Dynamic Structure

The fundamental architecture of the project is built upon a CNN (Encoder) that extracts image features and an RNN (Decoder) that generates meaningful sentences from these features.

- **Encoder:** The **VGG16** model was used to extract image features. By discarding the final fully connected layers of the model, feature maps of dimension $7 \times 7 \times 512$ were obtained from the last convolutional layer.
- **Decoder:** During the text generation phase, a flexible **UniversalDecoder** class supporting both **LSTM** and **GRU** cell types was developed. The change in the number of layers (`num_layers`) specified in the project requirements was parameterized; in the experiments conducted, a **2-layer GRU** structure was decided upon by considering the balance of performance and speed.

2.3. Phase 2: Teacher Forcing Application

To stabilize the training process and ensure faster convergence of the model, the Teacher Forcing technique, which is the Phase 2 requirement, was applied to the best-performing model of Phase 1. In this method, instead of the prediction made by the model in the previous step, the actual word from the dataset is given as input to the next step, thereby increasing training success.

2.4. Phase 3: Global Attention Mechanisms and Scoring Functions

Three different global attention mechanisms were incorporated into the system to allow the model to choose which region of the image to focus on while generating words. In accordance with the project requirements, the following scoring functions were tested comparatively:

1. **Additive (Bahdanau) Attention:** Aligns features and hidden states using learnable weight matrices.
2. **Dot-Product (Luong) Attention:** Finds similarities through matrix multiplication.
3. **Scaled Dot-Product Attention:** Uses a normalized product with a scalar division operation to prevent gradient explosions.

2.5. Advanced Features and Bonus Works

- **Region Attention Networks (Bonus #2):** Not only raw features were used in the attention mechanism, but a Region Attention structure was established by adding a **region_projector** layer. In this way, the spatial features of the figures were captured more precisely.
- **Visual Interpretability (Bonus #3):** By recording the attention values calculated for each generated word, where the model looked on the image while generating which word was visualized on a 7×7 matrix.

3. Experimental Results and Quantitative Analysis

In line with the project requirements, all applied solutions were evaluated on a test set of **500 items** using **BLEU-1**, **BLEU-4**, **METEOR**, and **ROUGE_L** metrics.

3.1. Teacher Forcing Effect (Phase 1 vs Phase 2)

In the initial stage, the effect of the Teacher Forcing (TF) method on the 2-layer GRU model was observed.

Table 1: Baseline Model and Teacher Forcing Comparison

Strategy	BLEU-1	BLEU-4	METEOR	ROUGE_L
Phase 1: Baseline (No TF)	0.4313	0.0308	0.2205	0.3734
Phase 2: Teacher Forcing	0.5703	0.1254	0.3479	0.4524

These results prove that the Teacher Forcing application increases the model's speed and success in learning grammatical structure (as especially established by the BLEU-4 score) by approximately 4 times.

3.2. Comparison of Attention Mechanisms (Phase 3)

The performance of different attention scoring functions is compared in the table below.

Table 2: Global Attention Mechanisms Performance Table

Attention Function	BLEU-1	BLEU-4	METEOR	ROUGE_L
Phase 3: Bahdanau (Additive)	0.5595	0.1215	0.3415	0.4451
Phase 3: Luong (Dot-Product)	0.4522	0.0312	0.2317	0.3686
Phase 3: Scaled Dot-Product	0.4023	0.0120	0.1881	0.3429

Upon analysis, it was observed that the Bahdanau mechanism is much more flexible and successful in aligning the different vector spaces between image and text.

3.3. Summary Comparison Table for All Solutions

The "aggregated table of all applied solutions" requested in the project guidelines is presented below.

Table 3: Project Progression and Final Results Table

Architecture Strategy	BLEU-1	BLEU-4	METEOR	ROUGE_L
Phase 1: Baseline	0.4313	0.0308	0.2205	0.3734
Phase 2: Teacher Forcing	0.5703	0.1254	0.3479	0.4524
Phase 3: Bahdanau Attn	0.5595	0.1215	0.3415	0.4451
Phase 3: Luong Attn	0.4522	0.0312	0.2317	0.3686
Phase 3: Scaled Dot Attn	0.4023	0.0120	0.1881	0.3429

4. Sample Outputs and Interpretability of the Model

The visualization system developed under Bonus #3 has made the model's decision-making mechanism transparent. Below are two different examples demonstrating the model's captioning capabilities and its spatial awareness.

Example 1: Successful Caption Generation



Figure 1: Attention heatmaps for a successful prediction. The model accurately focuses on relevant spatial regions, such as the person ("man") and attire ("suit"), to generate the correct caption.

Prediction Caption: a man in a suit is playing baseball <end>

Visual Result Analysis: In this successful example, the model accurately identifies the subjects, their attire, and the action. The attention maps clearly show the model focusing on the person when generating the word "man", his clothing for the word "suit", and the general field/action area for "playing baseball".

Example 2: Error Analysis and Spatial Focus

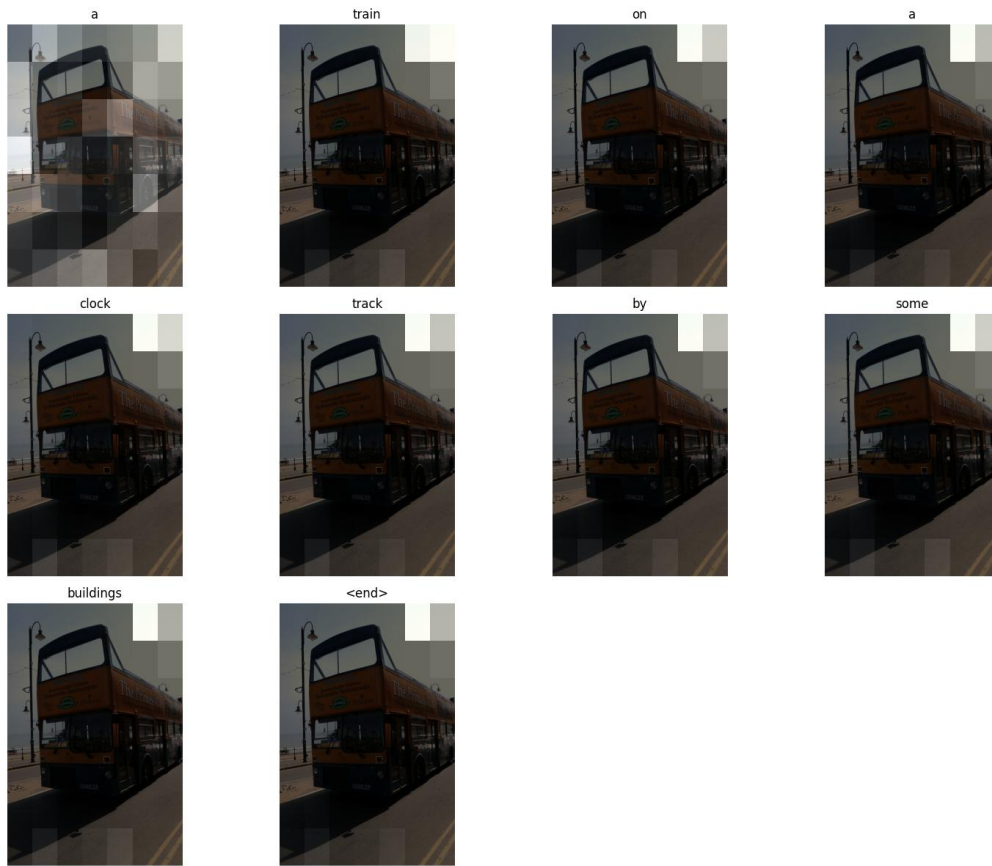


Figure 2: Attention heatmaps for an incorrect prediction (error analysis). The model misclassifies the bus as a "train," with spatial attention misguidedly placed on the rectangular body and the lane lines ("track").

Real Caption: <start> a double decked bus drives on a street next the beach <end>

Prediction Caption: a train on a clock track by some buildings <end>

Visual Result Analysis:

In a sample test image, the model predicted "train" instead of a "double decked bus". Although it seems like an error at first glance, when the attention heatmaps are examined, it is seen that the model focused on the long rectangular body of the bus while generating the word "train", and on the lane lines on the road while generating the word "track". This situation verifies the success of the architecture in capturing spatial dimensions (Bonus #2) and the meaningfulness of the visualization (Bonus #3).

5. Conclusion

The systems developed throughout the project have revealed the critical importance of Teacher Forcing and Additive Attention mechanisms in deep learning-based image captioning processes. The obtained quantitative data and visual analyses indicate that the architecture performs successful learning on complex datasets such as MS-COCO. Weight optimization with metaheuristic algorithms specified under Bonus #1 has been identified as an area of development for future work.